

Intelligent Agents

Lecture 1

Introduction to Agents

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
- An agent gets percepts one at a time, and maps this percept sequence to actions.
- What are percepts ?
 - Percepts are the electrical signals from the sensors after processing objects in visual field. (like location, color, direction, loudness, etc.)

Introduction to Agents

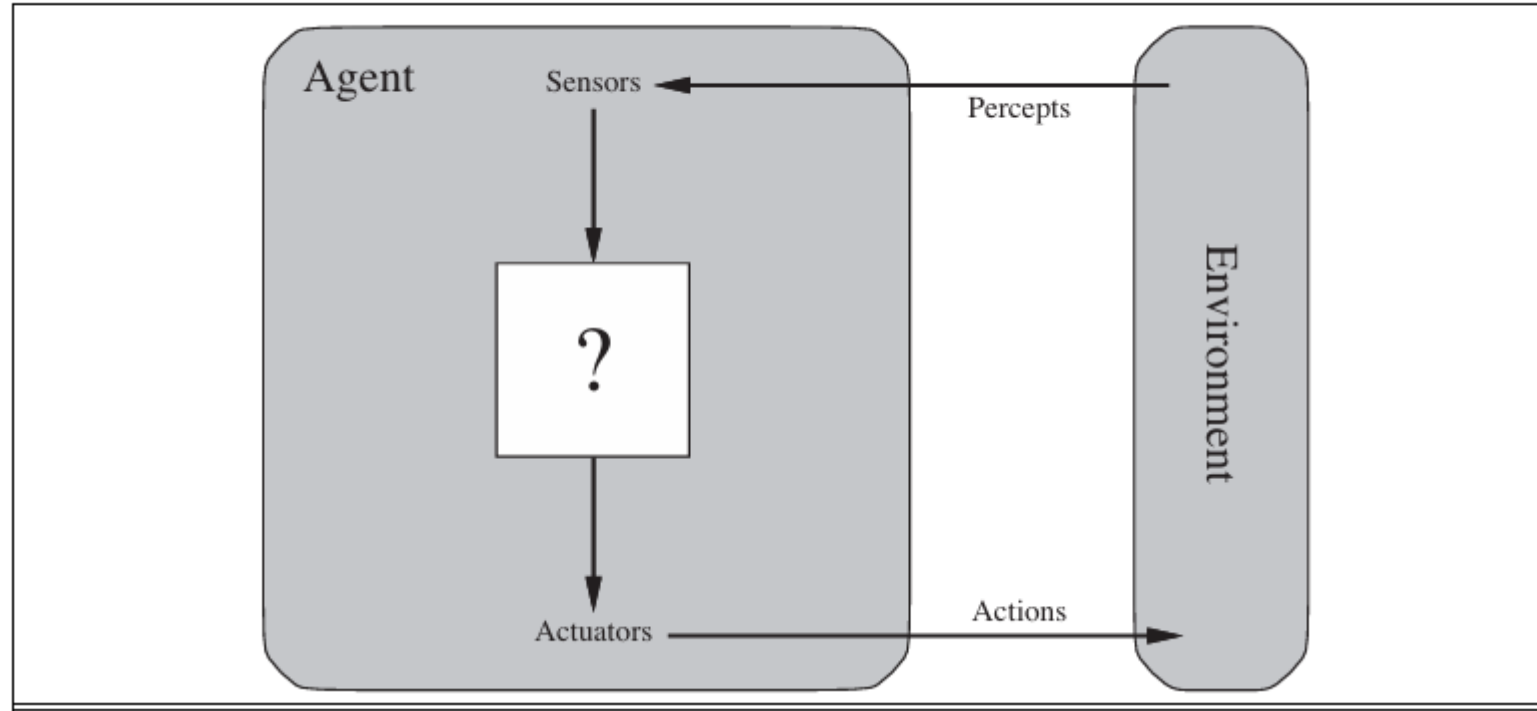


Fig: Agents interacting with the environment through sensors and actuators.

Examples of agents

- For a human
 - Sensors: Eyes (Vision), Ears (hearing), skin(touch), tongue(taste), nose(smell)
 - Percepts:
 - At lowest level: Electrical signals from these sensors
 - After preprocessing- objects in the visual fields (location, textures, colors), auditory streams(pitch, loudness, direction)
 - Effectors(Actuators) : Limbs, Digits, Eyes, Tongue
 - Actions: Lift a finger, Turn left, walk, run

Automated Taxi Driving System

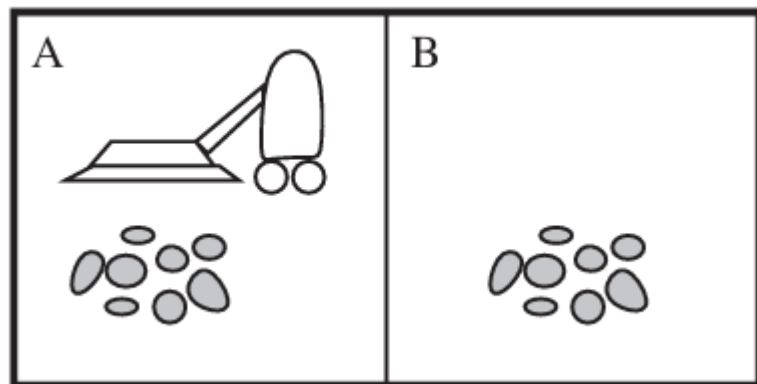
- Percepts: Video, sonar, speedometer, odometer, engine sensors, keyboard input, microphone, GPS,
- Actions: Steer, accelerate, brake, horn, speak/display
- Goals: Maintain safety, reach destination, maximize profits(fuel, tire wear), comply to the laws, provide passenger comforts,
- Environment: Urban streets, freeways, traffic, pedestrians, weather, customers,

Few examples of Agents

- Automatic Streetlight System
 - Percepts:
 - Actions:
 - Goals:
 - Environment:
- Traffic Signal Control System
 - Percepts:
 - Actions:
 - Goals:
 - Environments:

Agents

- Percept refers to agent's perceptual input at any given instant
- Percept sequence is complete history of everything the agent has ever perceived.
- Mathematically, agent's behavior is described by the agent function that maps any given percept sequence to an action.
- Let us take an example of vacuum cleaner world as shown in the next slide. For simplicity, we consider the particular world has just two locations: A and B. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.



Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Properties of Agent

- Autonomous
 - An agent can operate on its own without constant human guidance. It makes decisions independently based on its internal state and knowledge.
- Interacts with other agents plus the environment
 - An agent communicates or collaborates with other agents and perceives/acts upon its surrounding environment to achieve tasks.
- Reactive to the environment
 - An agent can sense changes in its environment and respond promptly. It adapts its behavior based on new incoming information.
- Pro-active (goal directed)
 - An agent does not just react—it also takes initiative. It sets goals and works strategically toward achieving them, planning ahead when necessary.

Rational Agent

- How to measure rationality ?? What makes an agent rational ??
- Consider, for example, the vacuum-cleaner agent from the preceding section. We might propose to measure performance by the amount of dirt cleaned up in a single eight-hour shift. With a rational agent, of course, what you ask for is what you get. A rational agent can maximize this performance measure by cleaning up the dirt, then dumping it all on the floor, then cleaning it up again, and so on.
- A more suitable performance measure would reward the agent for the clean floor.
- The bigger picture here is that “it is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.”

Rational Agent

- What is rational at any given time depends on following four things:
 - The performance measure that defines the criterion of success.
 - The agent's prior knowledge of the environment.
 - The actions that the agent can perform.
 - The agent's percept sequence to date.
- Thus, rational agent can be defined as *“For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.”*

Intelligent Agents

Lecture 2

Task Environment

- In our discussion of rationality of simple vacuum agent, we had to specify performance measure, environment and the agent's actuators and the sensors.
- They are grouped as PEAS in acronym.
- Task environment means PEAS description of the environment
 - Performance
 - Environment
 - Actuators
 - Sensors

Automated Taxi Driver System

- Performance Measure
 - We would like to include desirable qualities such as getting to the correct destination, minimizing fuel consumption and wear and tear, minimizing the trip time or the cost, minimizing the traffic violations, maximizing passenger safety and comfort, maximizing profits.
- Environment
 - What type of driving environment will the taxi face ?
 - A taxi must deal with variety of roads, ranging from rural lanes to 12 lane freeways.
 - The road contain other traffic, pedestrian, stray animals and other difficulties.
 - It must interact with potential and actual passenger too.

Automated Taxi Driving System

- Actuators

- The actuators for an automated taxi include those available to a human driver: control over the engine through the accelerator and control over steering and braking.
- It will need output to a display screen or voice synthesizer to talk back to the passengers, and perhaps some way to communicate with other vehicles .

- Sensors

- The basic sensors for the taxi will include one or more controllable video cameras to focus on the road.
- It might augment these with infrared or sonar sensors to detect distances to other cars and obstacles.

Agent types and PEAS descriptions

- Medical Diagnosis System
 - P
 - E
 - A
 - S
- Satellite Image analysis system
- Part picking robot
- Refinery controller
- Interactive English tutor

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

Types of environments

- Fully observable vs partially observable
 - If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable.
 - It is important because the agent need not maintain any internal state to keep track of the world.
 - An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
 - For example, a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking.
 - If agent has no sensors at all, then it is unobservable.

Single agent vs multiagent

- A single agent environment is one where only one agent is trying to achieve its goal, and other entities do not actively interfere or compete.
- Only one intelligent agent is acting and environment may be complex but no competitors or collaborators.
- Multiagent environment involves two or more intelligent agents whose actions effect each other's outcomes.
- Multiple agents may compete or collaborate with each other.
- For example:
 - Single agent: Vacuum cleaner robot (simple model), crossword puzzle solving
 - Multiagent: Self-driving cars at intersection (co-operate with each other)
 - Soccer Robots (Teams try to score against each other)

Deterministic vs Stochastic

- If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic.
- If the environment is partially observable, however, then it could appear to be stochastic.
- Most real-world situations are complex, and have indeterministic properties and hence mostly are stochastic.
- If we consider taxi driving case then one can never predict the behavior of the traffic and there may be cases like tire burst and one's engine seizes up without warning.

Episodic vs Sequential

- In an episodic task environment, the agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes.
- Most of the classification tasks are episodic.
- In sequential environment, on the other hand, the current decision could affect all future decisions. Chess and taxi driving are sequential: in both cases, short-term actions can have long-term consequences.

Static vs Dynamic

- If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise, it is static.
- Static environments are easy to deal with because the agent need not keep looking at the world while it is deciding on an action, nor need it worry about the passage of time.
- Dynamic environments continuously ask the agent what it wants to do.
- If the environment itself does not change but the agent's performance score does, then, environment is semi-dynamic.
- For example:
 - Static: Crossword puzzles
 - Dynamic: Automated Taxi Driving
 - Semi-Dynamic: Chess, when played with clock, timed exams, auction bidding system

Discrete vs Continuous

- The discrete/continuous distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent.
- For example, playing chess has discrete number of states whereas taxi driving has continuous state and continuous-state problem.
- The speed and location of taxi move through a continuous values and vary smoothly over time and hence, termed as “continuous”.

Known vs Unknown

- In a known environment, the outcomes for all actions are given.
- The agent fully knows about:
 - The rules of environment
 - How its actions affect the world
 - The consequence of each action
 - What possible states and rewards are
- In an unknown environment, the agents do not know about:
 - The full rules
 - Consequence of action
 - Structure of environment
 - Reward model
- The agent must learn through trial and error, and exploration about how the environment works.
- For example:
 - Known environment: Chess, mathematical puzzle solver
 - Unknown environment: Self-driving cars in new city, a robot exploring mars terrains

Self-driving car

- Observability:
- Single/Multi agent
- Deterministic/Stochastic
- Episodic/Sequential
- Static/Dynamic
- Known/Unknown

Washing Machine

- Observable:
- Single/Multi agent
- Deterministic/Stochastic
- Episodic/Sequential
- Static/Dynamic
- Known/Unknown

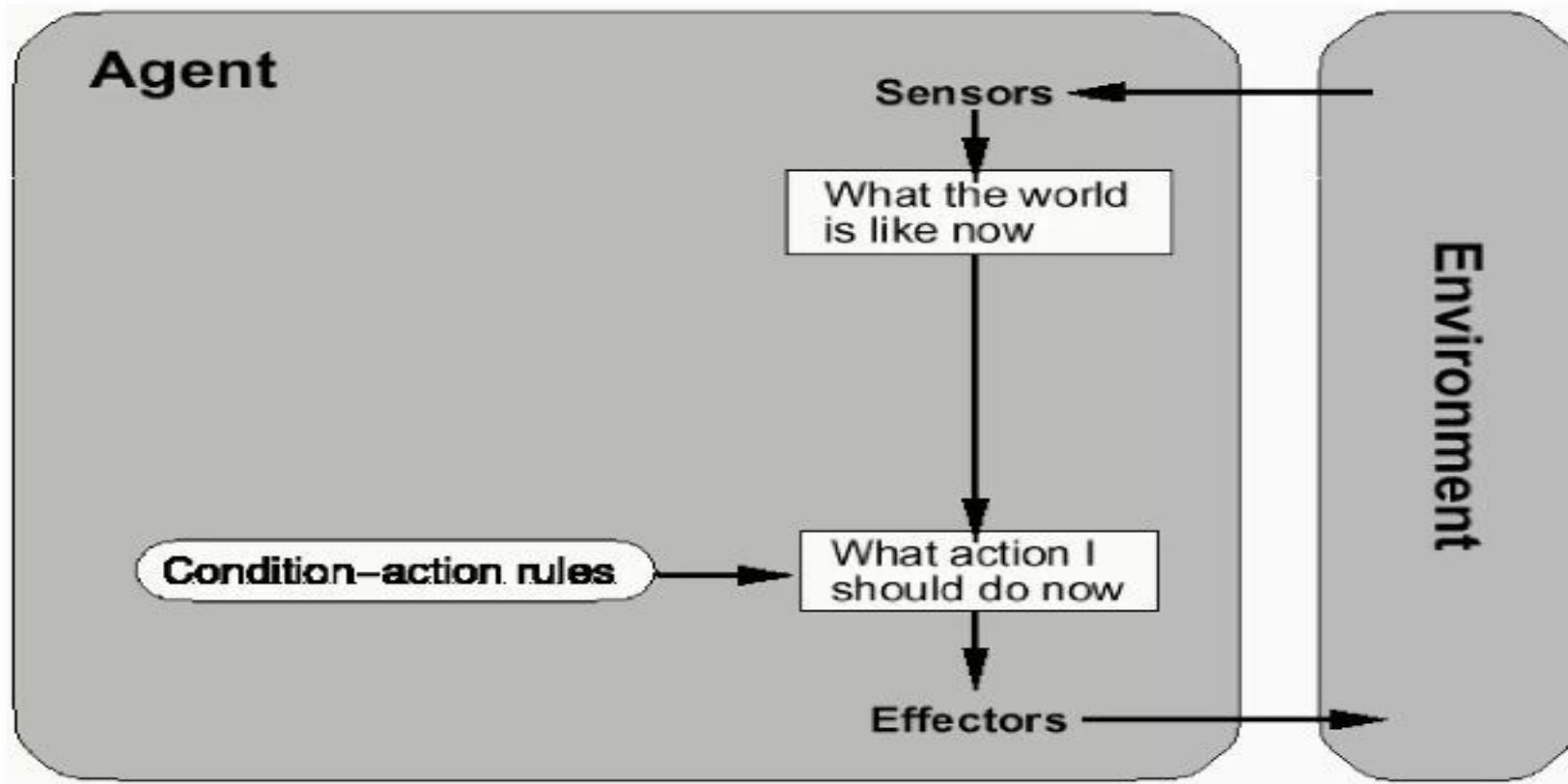
Types of Agents

- Simple reflex agents
- Model-based reflex agents
- Goal based agents
- Utility based agents
- Learning Agents

Simple Reflex Agents

- Simple reflex agents act only on the basis of the current percept, ignoring the rest of the percept history.
- Environment is fully observable, deterministic, static, episodic, discrete and single agent.
- The agent function is based on the condition action rule :
 - If condition $\rightarrow$ then action:
 - For example, If (you see car in front's brake lights) then apply brakes.
 - This works only in certain types of cars, but unfortunately won't work in older configurations of taillights.
 - A simple reflex agent in such case would either brake continuously or not even brake at all.
 - So, even a little bit of unobservability can cause serious trouble.

Simple Reflex Agents



Model based reflex agents

- The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now.
- That is, the agent should maintain some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
- This knowledge about “how the world works” is called model of the world.
- An agent that uses such a model is called a model-based agent.

Model based reflex agents

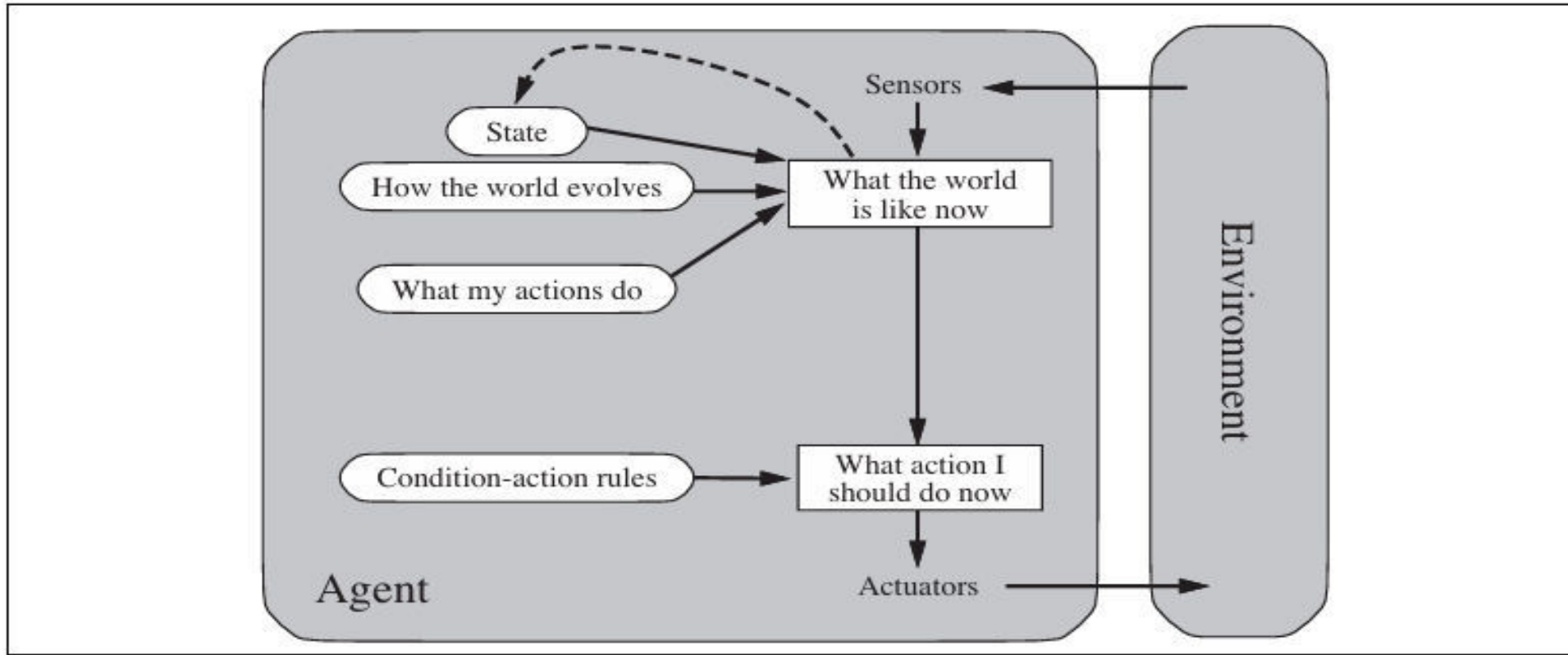
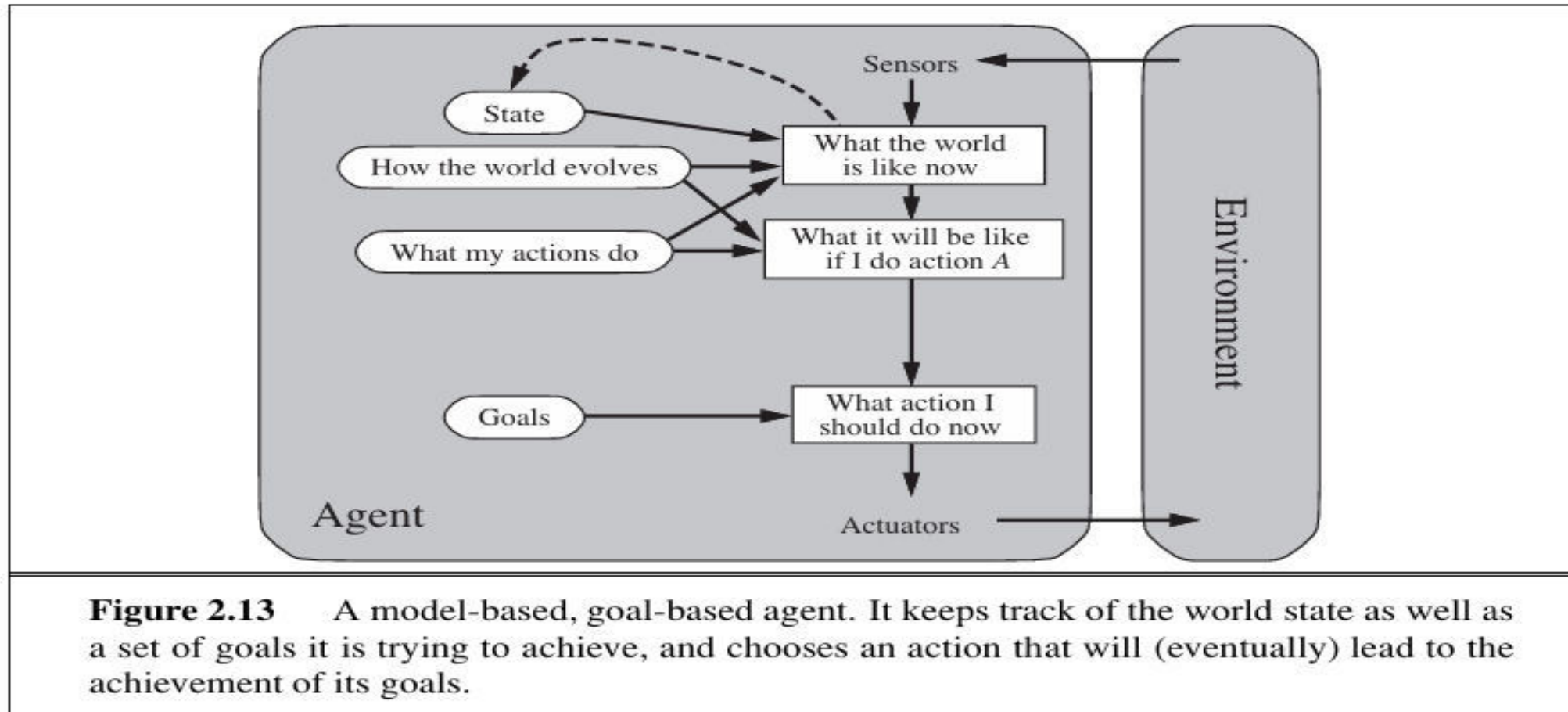


Figure 2.11 A model-based reflex agent.

Goal based agents

- Knowing something about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, right or straight on.
- The correct decision depends on where the taxi is trying to get to. In other words, as well as a current state description, the agent needs some sort of goal information that describes situations that are desirable—for example, being at the passenger's destination.
- The agents can combine this goal information with the model to choose actions that achieve the goal.
- Goal based agents can be straightforward-when the goal satisfaction results from an immediate single action but can be tricky when we have to consider a longer sequences.
- Goal based agents may appear to be less significant but they are more flexible as the goals can always be modified.

Goal based agents



Utility based agents

- Utility based agents select their actions based on the utility values. i.e. how good an outcome is.
- Goals are binary i.e. whether the actions are acceptable/unacceptable or gets job done/does not.
- So, utility goes beyond goals maximizes happiness/satisfaction, not just reaching a goal.
- For example, a self driving car does not just want to reach to the destination. It wants to reach it in the best possible way, balancing safety, time, comfort, fuel efficiency, and passenger satisfaction.

Utility based agents

- In a case like self-driving car, utility function can be calculated in this way:
 - $Utility = f(\text{safety, travel time, comfort, fuel use, legal constraints})$
 - More safety -> higher utility
 - Less time -> higher utility
 - And so on.
- A simple utility function can look like this:
 - $Utility = w1 * \text{Safety} + w2 * \text{Comfort} + w3 * \text{Speed} + w4 * \text{Efficiency} - w5 * \text{RuleViolations}$
 - $w1, w2, w3$ are weights showing how important each factor is.
- How to choose the next action in a car ?
 - Following are the actions and their expected utility:
 - Stay behind = 82
 - Overtake = 74
 - Change lane = 88
- The car chooses “Change lane” in this case because it has the highest utility.

Utility based Agents

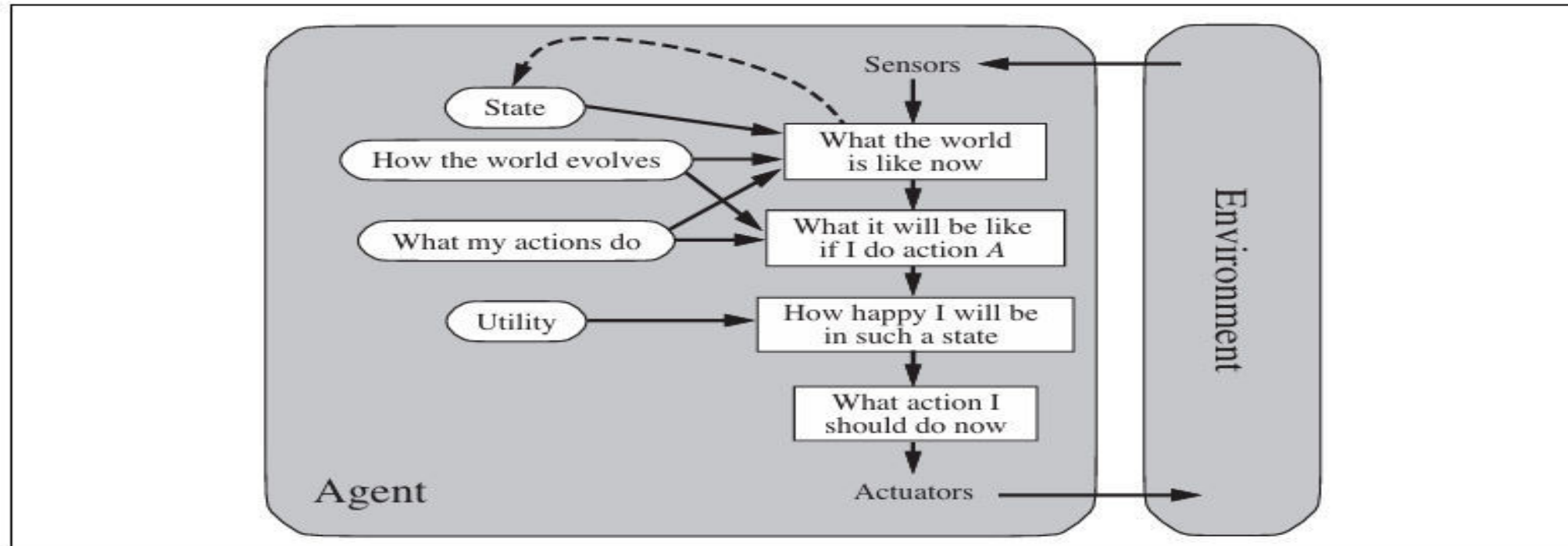


Figure 2.14 A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

Learning Agents

- An agent that improves its performance over time based on experience.
- Learns from interactions with the environment instead of relying only on fixed rules.
- Learning helps the agents handle new, unseen situations effectively.
- Components of Learning Agent
 - Performance Element
 - The part that actually selects actions.
 - Uses the current policy to decide what to do.
 - Learning Element
 - Responsible for improving the agent's behavior.
 - Analyzes the feedback and modifies the performance element

Learning Agents

- Components of Learning Agents
 - Critic
 - Evaluates the actions taken by the agent.
 - Provides feedback like reward, score or error signals.
 - Problem generator
 - Suggests exploratory actions.
 - Helps the agent try new things to discover better strategies.
- How does it learn?
 - Observes its environment
 - Takes an action
 - Receives reward or feedback indicating how good the action was.
 - Adjust its decision-making process to increase rewards in future.
 - Repeat this cycle and become more effective.

Learning Agent

